NETWORKING AND SECURITY OPERATIONS WITH SIEM, FORENSICS AND TRAFFIC ANALYSIS

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Date : 28 February 2026

System : Ubuntu (VM), Kali Linux (VM), Metasploitable 2 (VM), Windows 11 (Host)

Introduction :

- Modern organizations rely heavily on secure and monitored network infrastructures.
- This project focuses on designing a secure small-scale network, analyzing traffic patterns, configuring a SIEM system, simulating an attack using Metasploit, and conducting network forensics and threat hunting.
- The objective is to understand real-world cybersecurity operations are given below.

Objectives :

- Subnetting.
- Traffic monitoring.
- Protocol troubleshooting.
- Log monitoring.
- Incident simulation.

- Digital forensics.
- Threat hunting.

Network Design and Subnetting :

IP Address Planning :

- Base Network : 192.168.10.0/24
- Requirement :
  - — 1 Router.
  - — 1 Server.
  - — 10 Client Systems.

Subnet Used :

- Subnet Used : 192.168.10.0/28

Subnet Details :

Item	Value
Network ID	192.168.10.0
Subnet Mask	255.255.255.240
Total Hosts	16
Usable Hosts	14
Broadcast	192.168.10.15

Network Diagram Components :

- Router – Firewall – Switch.
- Switch connected to :
  - — Kali Linux attacker machine.

— Windows victim machine.

— Log server (ELK).

- **Tools Used :**

— VirtualBox.

— Kali Linux.

— Windows VM.

Traffic Analysis Using Packet Capture :

- **Tool Used :** Wireshark.

Observed Traffic :

— TCP 3-way handshake.

— DNS queries.

— HTTP requests.

— ICMP ping traffic.

— Suspicious FTP traffic.

Analysis :

Packet inspection revealed :

— Source IP initiating scan.

— Destination port patterns.

— Multiple SYN packets indicating scanning activity.

— Bar charts and packet snapshots were captured and analysed.

— Performed the scanning using filters.

— Significant activity level.

Protocol Troubleshooting :

- DNS resolution failure.
- Incorrect gateway configuration.
- Blocked port 21 (FTP).

Troubleshooting steps :

— Used ping.

— Used ipconfig.

— Used tracert.

— Checked firewall rules.

Problem was resolved by correcting gateway configuration and opening required port.

SIEM Setup Using ELK Stack :

Tool Used :

— Elasticsearch.

— Logstash.

— Kibana.

Configuration Steps :

— Installed ELK on Linux server.

— Configured log ingestion.

— Enabled system logs and network logs.

— Created dashboards in kibana.

Log Monitoring :

Dashboards displayed event spikes during simulated attack.

— Authentication attempts.

— Port scan logs.

— Failed login attempts.

— Suspicious IP activity.

Attack Simulation Using Metasploit :

Tool Used :

— Metasploit.

— Attack VM : Kali Linux.

Exploit Used :

- use exploit/unix/ftp/vsftpd_234_backdoor

— The exploit targeted vulnerable FTP service.

After exploitation :

— Interactive shell obtained.

— Unauthorized access simulated.

Network Forensics Investigation :

Evidence Collected :

— Packet capture (PCAP file).

— SIEM logs.

— System logs.

— IP source tracing.

Findings :

— Attacker IP identified.

— Port 21 targeted.

— Multiple SYN packets before exploitation.

— Backdoor triggered via FTP service.

— Timeline reconstruction performed using logs.

Threat Hunting :

Threat hunting was conducted using following services.

- Log correlation in kibana.
- Suspicious IP filtering.
- Detection of abnormal traffic volume.
- Checking repeated authentication failures.

Indicators of Compromise (IoCs) :

— Attacker IP address.

— Exploit pattern signatures.

— Abnormal port scan behaviour.

Security Recommendations :

1. Disable vulnerable FTP versions.
2. Apply regular patch updates.
3. Implement firewall port restrictions.
4. Enable IDS/IPS.
5. Continuous SIEM monitoring.

6. Use strong authentication policies.

Conclusion :

- The simulation reflects real-world SOC (Security Operations Center) workflow, from detection to investigation and mitigation.
- The hands-on experience enhanced understanding of network defense mechanisms and incident response methodology.

This project successfully demonstrated :

1. Secure subnet design.
2. Traffic analysis using Wireshark.
3. Protocol troubleshooting.
4. SIEM implementation using ELK Stack.
5. Exploitation using Metasploit.
6. Network forensic investigation.
7. Threat hunting practices.

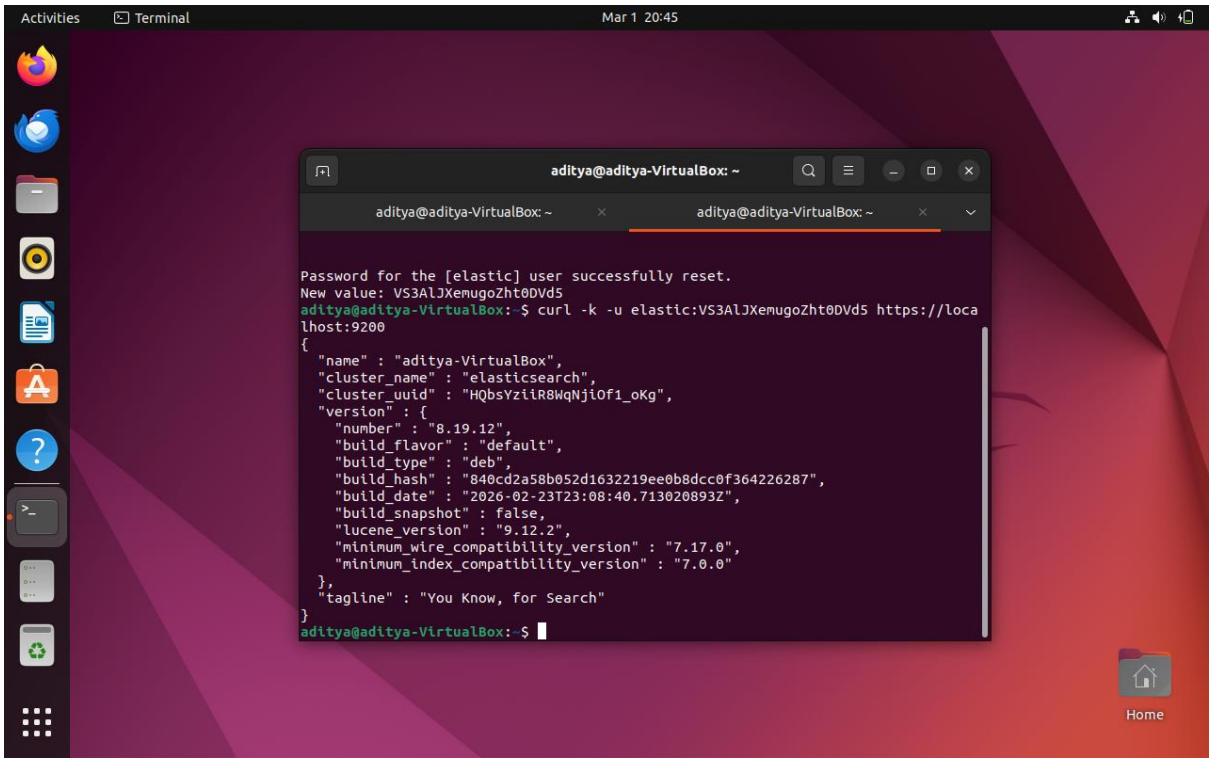
Key Learnings :

- Gained practical understanding of subnetting and CIDR allocation.
- Learned real-time packet inspection using Wireshark.
- Understood log ingestion and monitoring using ELK Stack.
- Experienced vulnerability exploitation using Metasploit.
- Performed basic network forensics and timeline reconstruction.
- Identified Indicators of Compromise (IoCs) through threat hunting.
- Improved troubleshooting of network and protocol issues.

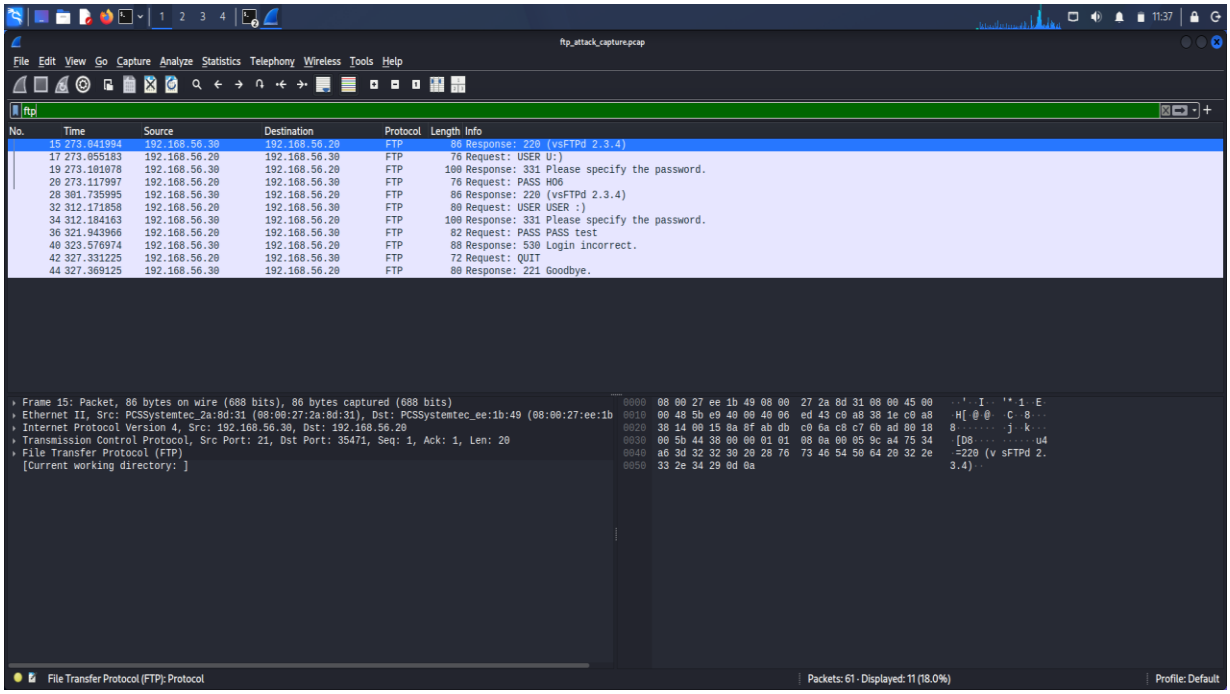
Challenges Faced :

- Performance issues due to limited 8GB RAM while running VMs.
- Configuration errors in ELK pipeline setup.
- Payload configuration confusion in Metasploit.
- Correlating Wireshark packets with SIEM logs.
- Managing network connectivity between virtual machines.
- Understanding exploit behaviour in controlled environment.

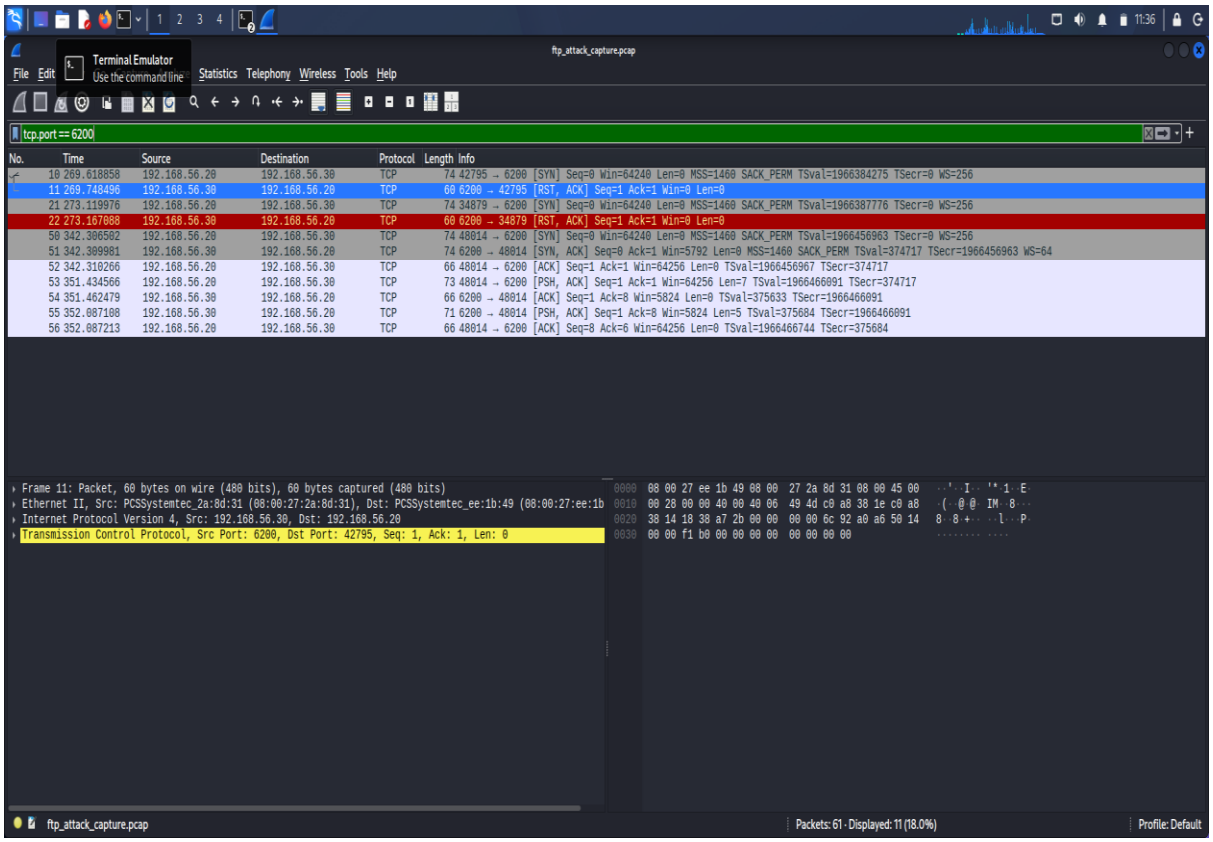
Snapshots & Evidence :



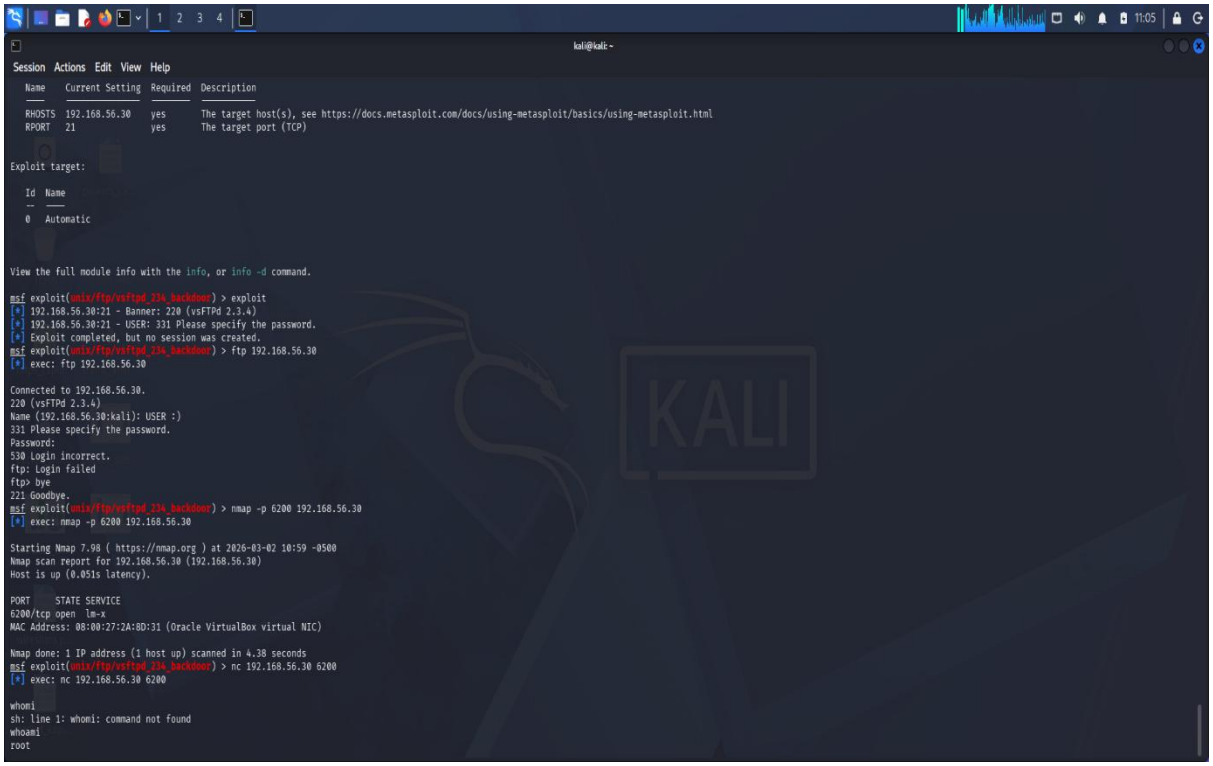
[Elasticsearch Configured]



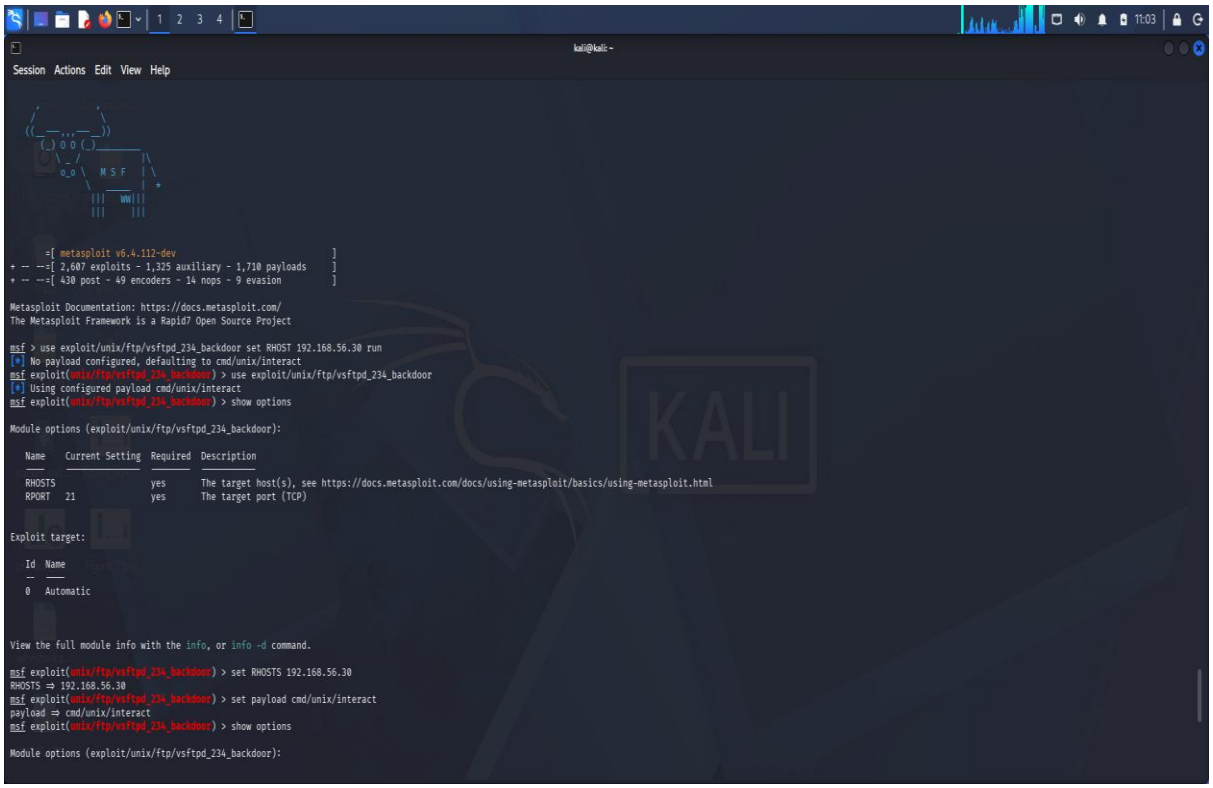
[FTP traffic analysis – Wireshark]



[TCP port 6200 traffic captured – Wireshark]



[Exploiting Target – Kali, Metasploit]



[Payload Configured during Exploitation]